ActInf MathStream 011.1 ~ "Structured Active Inference", Toby St Clere Sm

MathStream | Toby St Clere Smithe | 1:29:53

Hello and welcome. It is August 12th, 2024, and we are in ActInf Math Stream number 11.11111111

and with repeat friend Toby Smythe. Today we're going to talk about structured active inference.

There will be a presentation followed by a discussion. So thank you Toby for joining and everybody for watching and asking questions and looking forward to learning more here. So go for it.

Cool. Great. Thanks, Daniel. Thanks for having me on again. So as Daniel says, this is going to be a talk about structured active inference, which is really my account of the structure of active inference systems.

And really it kind of extends beyond active inference to any kind of system that you could think of as being kind of agential and that has some kind of compositional structure in particular.

But we'll get to that. Okay. The way I like to begin thinking about structure and active inference is at the moment it's kind of like,

if you wanted to model in active inference, an agent or a person getting onto a bicycle or getting into a car,

then I think as things stand, it might be a bit tricky because that process involves like changing the agent's Markov blanket or how I like to think of it is changing the agent's interface.

And to be able to talk about that precisely, you need to have a good notion of what interface is.

So, you know, you, as you're, you know, getting into the, getting onto the bike, let's say, you move, you change from, you know,

walking by taking steps one after another to, you know, like changing your direction by like leaning and pedaling.

So the kind of type of actions you can take and the things that you pay attention to change, yes,

because you're now this kind of composite system, you and the bicycle.

And it's a similar story when you get into the car.

So that, that really, that means that you don't just have like one like set of observations and like one set of actions you can take.

This, this kind of composing process means that you have to have a bit more flexibility.

And in particular, you need to say precisely what those sets of observations and sets of actions should be.

So the basic idea behind structure, structured active inference is really just to put the ideas of active inference into categorical systems theory.

And so categorical systems theory is just an account of how dynamical systems of very general kind.

are made up of parts and how those parts compose together.

And it separates out the notion of the system from the notion of its interface.

And so this allows us to be very precise about what is the kind of agent Markov blanket and how does that change?

So we inherited a number of features of categorical systems theory in this notion of structured active inference.

In particular, we inherit this kind of idea of structured interface.

And that means not only do we have to be precise about what the interfaces are, but we also are able to say what the structure in some sense of the interface is and how does it change?

And how does it interact with other kinds of interface?

I'll say a lot more about that in the next few slides.

Because categorical systems theory is all about the composition of systems and their interfaces, so is structured active inference.

And that means that we get a nice kind of modular account of active inference.

And we can compare active inference with other accounts of agency.

And we can compare active inference systems of various different kinds in a nice precise way.

And so that means we could sort of translate between continuous time and discrete time systems or various different kinds of continuous time or stochastic systems.

Those are all different systems theories of their own.

And active inference has lots to say about how to do inference and policy selection on those different kinds of systems.

But categorical systems theory means that we're very precise about what those different systems and how that inference process works.

And so this kind of comparability comes into, is formalized by the notion of morphism of systems.

So these morphisms, they come into two kinds in systems theory.

We get morphisms of interface.

Those are the things that allow you to sort of plug yourself into another sort of bigger system in some sense.

That is, as you get onto the car, you get onto the bicycle, you're sort of wiring yourself into this composite human bicycle system.

But they also, these morphisms, they also encompass changing the model, that is changing the model of system, but also changing an individual model itself.

So doing things like structure learning, parameter change, so normal learning, meta learning, or things that we might think of in neuroscience as effective connectivity.

These are all kinds of morphisms of system.

And all of this stuff comes out of packaging active inference into this categorical systems theory language.

So with this, we gain a lot of things.

Not only do we sort of inherit these features, but we gain some nice new capabilities as well.

So I've already talked about what we get from structured interfaces.

So that means that we get systems where you can, like, the kinds of actions you take might depend on the situation or context or mode that you're in,

or that can act on very sort of precisely defined structured interfaces like computer APIs.

We get notions of agents that manage agents in a precise compositional way or sort of meta agents that can use active inference itself to change their structure.

There seems to be an interesting link between structure learning, as in within one model, and planning, as in choosing your sort of policy, your series of actions.

Both of those seem to be about composing sequences of morphisms.

And another thing that comes out of putting active inference into the language of categorical systems theory is that we can also make contact with categorical logic, which can also be applied to categorical systems theory.

And that means that we can do things like, or in principle, can do things like specify systems goals as logical predicates,

talk about contracts that might have to hold between agents or sort of on their behaviors that you might,

something you might care about if you're interested in safety of artificial intelligences.

And this logic, because it's in this systems theoretic framework, has to be compatible with the patterns of interaction of the systems.

Now, I'm not going to say very much about this, because there's really quite a big and open problem,

like spelling out all of the details here, but I've got a couple of slides at the end about it.

And in that same kind of vein, and similar to this notion of policies as kind of being sequences of morphisms,

structured active inference, because we give some kind of typing or context information to the actions that are available,

that means that we can do things like type checking in principle on policies.

And that's a kind of rudimentary way to verify the safety of a policy, i.e. just that it makes sense.

So the basic idea of categorical systems theory, this is the math stream, so I'm going to be quite mathematical here.

So there are two ingredients, as I mentioned.

One is that we have a category of interfaces, so that is just something very syntactic.

It says, okay, here are the interfaces and marker blankets of all my different systems.

And those interfaces form the object of this category.

And then here are all the ways that these interfaces could sort of link up or connect.

And these these wirings, they become the morphisms of this category of interfaces.

But there's no information in the category of interfaces about the systems themselves.

It's really just a sort of syntactic gadget for talking about how things connect up.

And so to attach systems to interfaces, we have what's known as an index category.

So we have for each possible interface, there's a whole category of systems that might somehow like fill that interface with a system.

So we think of the interfaces like a box.

And then the index category of systems over that category of interfaces fills that box with an actual system.

It sort of brings it to life.

And it being index category means that these categories of systems over interfaces have to be compatible with wiring up.

So if you have an interface, one box and another box side by side, and you connect them together using one of the morphisms,

you're also able to connect the systems in this way compatibly.

That is, you know, once you've plugged your boxes together, the systems sort of interact in this way accordingly.

So that allows us to do this kind of connection of systems.

It's not normally just a category.

We also typically have what's known as the monoidal structure on the category of interfaces.

And that means that we can place systems side by side.

We don't just plug systems like one into another.

We can have like two systems next to each other and then sort of connect them into a bigger system.

That is like, you know, your body is made of lots and lots of cells next to each other, but they're all sort of interconnected in parallel.

And that's sort of formalized by this monoidal structure, which is sometimes called tensor and represented with this symbol here.

And so the systems theory, again, has to be compatible with this tensor, which means if you put two boxes side by side and systems in them,

then you also get a system in the sort of composite, like sort of double box system.

And so to do wiring, really often what happens is you have one box.

I've written it as P here and another box written as P prime.

You put them together into this P tensor P prime.

That means that you can connect these, a system over this and a system over this into a system over the joint interface.

And then say you've got some some wiring.

So wiring would be amorphism P tensor P prime to Q.

Then the system theory allows you to take that wiring and wire the systems together accordingly.

And then that means you've got a system on this like Q interface.

And so often you see diagrams and systems theory, which are a bit like this.

This is P, this is P prime, this is Q.

And then this is like some like way of wiring the systems together.

And then this is, you know, this is your wiring.

And then you've got the systems and you get a sort of composite system like this.

So I've got very fast.

This is the basics of categorical systems theory.

As I say, it's a language for talking about building complex systems from their parts in a nice sort of formal compositional mathematical way.

And so what we're going to do is put active inference into this language in a way that generalizes active inference in this sort of quite big new direction that allows for things like mode dependence of actions.

And so to do that, I want to introduce a particular category of interfaces, the category of polynomial functors.

If you know the work of David Spivak and his colleagues, he's a big proponent of this category for talking about things like systems and dynamics.

And so I'm going to adopt his notation in talking about this category.

He calls it poly.

And there's a nice book about this by David and Nelson New that you can read.

It's freely available online.

So a polynomial, it's really, again, as I say, it's like a syntactic gadget to talk about the interface of a system.

And it has kind of two key ingredients.

And it's really just like a polynomial that you see in high school algebra.

So it has like a set of summands, which are these like y to the somethings.

And they're indexed by this p of one set.

This is a set.

And each summand has another set attached to it, which is the exponent.

So it's really just like y squared plus 3y plus one.

This is a polynomial.

And this could be like sum over i, y to the xi.

If we think of the three y as like y plus y plus y.

And so this is a sort of general representation of a polynomial.

And we think of the indexing set p of one as the set of the system's outputs or the configurations that it could adopt or an active inference language like the observable states of a generative model.

And then to each possible configuration of the system, as I say, you have this set which represents the inputs or the actions that a system might take in that corresponding configuration.

And so that's just what how we think of a polynomial.

And then there are some things you can do with polynomials.

In particular, there's one like trivial polynomial, one that has just sort of one configuration and one possible input.

So there's no possible variation in its configurations and no interesting actions that it can take.

We can put polynomials side by side using this tensor product.

And this has the kind of expected effect on the polynomials.

A configuration of a tensor is like a configuration of each part.

And an input is or an action is an action or an input of the corresponding part like that.

And mathematically, we can translate between this kind of polynomial representation and what's known as a bundle, which is just a function.

So these are all sets, the P of I's and the P of one.

And so we can do the co-product or disjoint union of these sets by summing over the I's.

And this means that the elements of this big set here are pairs of an I, an indexing I in P of one and an element of the corresponding exponent set.

And this bundle is just a function that maps I X down to I.

And this is useful for defining the morphisms of polynomials, which will do the things that I think of as wiring.

And I'll talk about that in a moment.

But before I do so, I want to introduce one important example of polynomial, which are the monomials, because they don't have interesting.

They don't have many interesting sort of non-trivial summands.

They're just like one term.

So and we write these as O Y to the A.

And these are going to be familiar because they capture a kind of classical active inference.

So here I think of the O as like the set of observations of an agent and the exponent A as the set of actions.

And so these actions are like the inputs to a generative model that makes it transition.

And the observations are like the sort of data that the generative model or the environment produces that the agent receives.

And this is a polynomial because you're just summing over the A, the O's of each O, you have the same A.

And the bundle just takes a pair of O and A and projects down to O.

OK, cool.

So let me talk about how these things sort of wire up briefly.

This is done using the morphisms of polynomials, which are otherwise known as dependent lenses, if you know about functional programming, for instance.

So.

Amorphism is going to represent taking a system and putting inside a bigger system.

That is like if you've got two small boxes and you wire them together to bigger to build a big box or you take two cells and you put them or take multiple cells and you put them together to sort of make a multicellular organism.

So if we've got polynomials P and Q, morphism from P to Q is given by two things.

One sort of goes forward from P to Q and one sort of goes backwards.

And the forward one sort of takes the outputs of the inner systems and turns them into an output of the big system.

And the backwards one takes the inputs to the big system and the outputs of the inner systems and uses them to produce inputs to the small systems.

So like one of them sort of goes from inside the box to the out.

That's the the five one map.

And then the other one sort of goes in.

But it might take into account the sort of internal inputs as well.

So it can be used for like feedback.

So that is to say we've got this outputs map.

So we take outputs of the inner box to the outer box and for the corresponding inputs in the outer box.

So that's these wise.

We go backwards, but we have to account for this like mode dependence, which is a bit sort of funky.

But this just says that if we've got an output corresponding to position or configuration I of the inner box, then we're allowed to look at the inputs that correspond to the corresponding output.

And then we sort of back propagate those to the inputs again in configuration I.

So there's a bit of sort of accounting you have to do to keep track of these like the sort of of the configurations of all the systems.

But that just means that, you know, there's some like mode dependence or some like, you know, you might have some.

If you think of the sort of multicellular organism thing, then you might have some like, you know, ion channels which are open or closed on the on the membranes of the cells.

And what they do sort of depends on the state of all of those cells.

And so you have to account for that in these morphisms.

But in the monomial case where you don't have this like context or configuration dependence,

the morphisms become a little bit simpler.

They just become like a forwards thing.

So here we've got an A and an S going forwards.

And we have to figure out a way of giving an X.

And in the backwards direction, we have an A and an S on the inside and a Y as well coming in.

And we have to figure out a way of giving a B and a T.

So this really just encodes the wiring of these two boxes into the bigger box, like I said.

So using this category poly, whose morphisms are the dependent lenses and objects of the polynomials,

we can give a system theory which captures generative models.

I'm going to just talk about discrete time ones in the traditional sense on the monomials,

but also has some kind of more complicated mode dependent structure that I think could be useful in general as well.

And so this is going to be an index category, like I said.

It maps the polynomial, i.e. an interface of the kind that I was just talking about,

to a category whose objects are generative models on that interface.

So that means that they are going to be a quadruple like this.

So that is a state space, S.

I'm just going to think of it as a set.

A prior on the state space, which we could think of as like a goal or just prior beliefs about the agent states.

So that's a likelihood, which is a sort of stochastic output morphism.

In the case that this is just a monomial OY to the A, then that's just like P of O given S.

And then there's also this transition map, which says, OK, well, given that I am in this state,

I sort of can sample an input that has to correspond to the configuration that I'm in.

So I sample from my pi and then, OK, I'm allowed to receive an input in this corresponding configuration P of 1,

which is given by my likelihood.

And then, OK, I receive an input and I've got a state and I get a new state.

So that's like the transition function.

Again, in the sort of monomial case, this is exactly what we normally have in active inference.

We have a state S and an action in A and we get a new state.

It's just that now we're allowed to have the possible inputs for the system, i.e.

the possible actions that you could take on this generative model to be dependent on the state that the system or the state that the environment is in,

according to that model. And so that that captures this idea that, you know, say if I have my eyes closed,

I can't like receive visual signals anymore or, you know, if I open my eyes and I can.

So there's a difference in the sort of input signals that I can receive.

Or if I'm sort of plugged into my computer, I can't sort of easily get up and walk around because, you know,

I might like have to like unplug a wire or something.

So it captures this mode dependence.

There are sort of important sort of mathematical bookkeeping duties we have to do, which I'm not going to do,

but it's easy to check that it's like functorial, which means that it's indexed in the right way,

and that it's monoidal, which means we can put systems side by side.

And I've only just told you what the objects of this category, gen of P, are, but it also has morphisms.

And these are ways of like comparing systems that are known in computer science as co-algebra homomorphisms.

So this captures the notion of generative model that we're used to in active inference,

but it allows for a lot more structure, in particular, this kind of mode dependent structure and this compositional structure.

So just to sort of say like what kind of thing I mean by structured interface,

I'm taking an example from computer science because there are lots of uses of polynomial science in polynomials in computer science.

And one of them is for defining like domain-specific programming languages.

And I think that one use of this could be for defining like artificial agents that behave nicely with respect to sort of computer interfaces.

And so we can encode the data of a sort of domain-specific language into a polynomial,

because say we've got a little calculator language that has like two binary operations,

like plus and times that each take in an input, take in two inputs.

We've got a sort of unary operation that just takes in one thing and it negates it.

And like we've got some constants, like uncountably many that represent the real numbers,

then we can encode the data of this as like, you know, two y squared plus y plus r,

because we've got these two little operations, two binary operations.

They've got two possible inputs, one unary operation has got one input,

and like these constants which don't take in inputs.

And then we can, you know, represent the set of all terms of this language as like a set by applying the polynomial to like some x.

Those x terms, those x elements represent the inputs.

And then we can think, okay, well, what would be a machine that like takes in terms of my little domain-specific language and returns values?

It's something like r, y to the p of x, which is again another polynomial.

And you can do lots of things like this.

Basically, you can encode like any small programming language or any sort of abstract syntax tree into a polynomial.

It's not just examples from computer science that are relevant, but they have a lot of structure.

And so I wanted to show off those examples.

I think you could do a lot of things in biology as well using this language.

But it's kind of more sophisticated than we can do with just sort of a set of observations and a set of actions.

Another thing that I wanted to mention is that in this context, it's quite natural to think of the actions or that an agent takes as morphisms in a category.

And this is where this typing information or this typing idea comes from.

Because, you know, in categories, morphisms have to compose.

And so we can check that they are sort of compatible before we compose them.

And so to give you an idea of how this works, we can think of categories as like little state machines.

So here I've drawn a little category.

It's got objects x, y, and z.

And it's got these morphisms.

It's got two morphisms from x to y, f and f prime.

And, you know, we've got g from y to z.

And we can go along f and then along g to go from x to y to z.

And that takes us from x to z.

Or we can go, you know, f prime to z and blah, blah, blah.

But these are like steps or paths in a little space.

And so, you know, we go from x to y and then y to z.

But it doesn't sort of doesn't make sense.

You know, if we had another object over here that wasn't connected to z, we could go over here.

But then we wouldn't immediately be able to go over z.

That's to say, like, if we had an island or a peninsula or an island with just one bridge onto it,

we could go onto the island.

But then to go somewhere else, we'd have to go back along the bridge.

Sometimes the places you can go depend on where you are.

And this encodes that data into the notion of action.

And indeed, every category is itself can be turned into a polynomial.

The configurations of the polynomial are the objects of the category.

And then the sort of inputs to a system would be the morphisms of that category out of each object.

That is to say, all the steps you could take out of each object.

And then a policy would have to be like a composable sequence of morphisms.

Or, you know, in active inference, you might have something more stochastic.

So you have a belief over such sequences.

But because you have this typing information, it means you can check that your policies make sense.

Like, you don't try to go somewhere and then somewhere from that destination, that intermediate destination, you don't try to go somewhere that is not connected or take an action that doesn't make sense in some other way.

And so we can use this to start to think about, like, safety of policies using the logic that comes with category theory.

This isn't something that I've elaborated, but I think it's a sort of starting step to talk about safe active inference systems in a very rigorous way.

And I already mentioned that other kinds of important notion are also morphisms in structured active inference, like changes of interface or changes of structure.

And that means I think that we can rigorously apply active inference to itself to do things like structure learning or to choose how you, you know, like connect yourself into a bigger unit, like into an organization.

And on that note, let me tell you about how in this setting we talk about systems that manage systems.

This is a generalization or will become a generalization of what is known in the active inference literature as D4 hierarchical active inference.

And so the sort of formal gadget that we use to talk about this is known as the closure of the category of interfaces.

That is to say, it has what's known as an internal harm harm because morphism is otherwise known as homomorphism.

And the category of polynomials and also the category of lenses has one of these internal harm.

That is a way to refer to the category within itself.

And that internal harm has to be given a polynomial form.

And so the sort of configurations of this polynomial are from, say, from P to Q.

They are the ways of wiring P into Q, i.e. the set of lenses from P to Q.

And then, you know, we have to give as well exponents to provide a polynomial.

And these exponents are going to be all the sort of data that the inputs to these sort of P systems,

if we think of the P as like the inner systems that they require.

And so they require outputs from P and inputs from Q.

And that's exactly what this like input data is here.

So if we so we've got the polynomial that represents how P's connected to Q's.

And we can think about what generative models would be over such a polynomial.

That is just like systems that control the wiring of P's into Q's.

Or as I've done in this example, control the wiring of a P1 and a P2 into Q.

And so what would these systems be?

Well, they would observe the outputs of P1 and P2 and then the inputs of Q.

And they would update their states accordingly, having made these observations.

And then they would output ways to connect P1 and P2 into Q.

And that really is like what a manager does in an organization or sort of high level system does

in a sort of hierarchical structure.

And there's a sort of nice fact that I'll use later on that if you have a sort of trivial inner system,

i.e. just with this polynomial Y, then the generative models on this HOM polynomial

are just the same as generative models on the outer polynomial.

So we can use this to capture not only these hierarchical systems, but just kind of regular systems as well.

There's an interesting sort of side fact or reason why this is known as the closed structure

is that this is the same as something that's known in computer science as currying.
You might have seen this, but if you have a function that takes like inputs in A and B and produces C,
like this is maps A, B to like F of A and B, we could also write this as like a function that maps from A
to maps from B to C.
And that says, okay, well, this maps A to F of A something, which is a map from B to C.
So this is the same idea.
It says, okay, well, the like ways of like mapping into this HOM polynomial
are the same as like mappings of the joint systems.
It's an important structure that gets used a lot in this context.
Okay.
So I spent a lot of time talking about structured generative models.
In particular, I introduced this hierarchical structure and I talked about the structure of,
you know, mode dependence in an sort of abstract and general way.
But it was always about generative models.
And I haven't really talked much about like doing action selection or state inference,
which is of course a very important part of active inference.
And in some sense, it's kind of what active inference is all about.
But those processes fit very naturally into the same framework.
They're something that you attach to a generative model.
Active inference always starts from a generative model.
And then describes the process for providing a policy to control it.
And so an active inference agent, or I think an agent more generally,
is made up of two things.
It's made up of the generative model.
And then something that I call a controller,
which basically chooses the policy to provide to the generative model.
It's a way to sort of take this generative model, which is an open system,
and like sort of render it closed.
It sort of takes the observations that are produced by the generative model,
and then turns them into actions that can go into the generative model.
And so there's some sort of nice duality between the generative model and the controller.
And it's this duality that allows us to formalize active inference in a nice compositional way.
One way I sort of started to like to think about this,
and I don't really know much about this subject,
so I won't make any claims that it's rigorous or sensible,
but I think of the sort of generative model as somehow like the consciousness of the agent.
It represents how the agent perceives the world,
and then the policy part is it's sort of the subconscious processes,

which will drive the agent in that generative model.
I mean, this is just sort of my thinking.
I don't know if it's very sensible.
But as I say, we can fit all of this into the same framework
by kind of taking poly and this category of generative models
and the home structure and sort of putting them all together
into a sort of nice bundle or nice mix.
So I had this two category called GenPoly made up of all of these ingredients.
The objects are the interfaces, the polynomials.
You can think of them as generalized Markov blankets.
And then now it's a two category.
So instead of between one object and another having a set of morphisms,
now we have a whole category of morphisms.
So we have both one cells and two cells.
So it's a two-dimensional structure.
But the home category from P to Q is just the systems on the home interface.
So that is generative models on the structured interface
that sort of takes P systems and turns them into Q systems.
So they're sort of hierarchical manager systems.
And then the two cells are ways of comparing these systems.
And I haven't really talked much about that,
and they won't feature much in this talk.
And they don't feature in the paper either.
But they're important if you want to talk about things like structure learning or logic.
And then the composition of systems is, as I've been saying all along,
we take a system over one sort of home polynomial
and a system over another home polynomial,
and then we wire them together.
So then we have like a sort of high-level manager here
and a sort of lower-level manager,
and they talk to each other to produce what we could think of as like a hierarchical organization,
which takes sort of very low-level things and turns them into high-level things.
And yeah, so we're allowed to tensor polynomials,
and that makes this gen poly two category into monoidal two categories,
so we can put systems side by side in the way you'd expect.
And it means we can draw an active inference system
in this kind of string diagram way that category theory people like.
So the generative model is something like this.

It takes in nothing, and it produces p things.
That is, produces, well, I guess this is a bi-directional thing.
So it sort of is a morphism from a trivial object into p,
and it produces p outputs and takes p inputs.
And then the policy is like it's dual.
So it takes in p inputs, and it produces what the generative model expects as output.
And then as I may have mentioned that, no, I didn't mention yet,
that this, as I've said before, at least in the case of just a single system,
just on p, the hum from y to p is just the same as p, so that's good.
This captures what I was saying before about generative models on an interface.
And then the controller is the model from p to y.
That means it produces a sort of closed system when you compose it,
with p.
And so that is to capture this picture that I drew before,
where you have like a box, which is the generative model,
which is open, and then this controller, which sort of closes it off.
And that's encoded in this sort of formality.
Okay, and so now we've got this definition of the controller of a generative model
as like being dual to it.
We can sort of unwind or decompress what that means mathematically.
So the home polynomial from p to y has this particular form.
It's just something that you can work through if you write down that expression
and figure out what it means.
The outputs or configurations of this polynomial are what are known as sections of p.
So a section of this p taken as a bundle is a way of going from the outputs
and for each output giving an input that matches the corresponding configuration.
So it's just a way of sort of making this diagram commute,
as we would say in category theory.
That is to say, okay, we're in an output or a configuration
and we say, okay, well, I need to provide an input.
And yes, that's nice.
It matches what configuration I'm in.
And we think of this as like being the data that the world supplies to the system.
So the system or like we think of the system as producing outputs,
the environment takes those outputs and then turns them into inputs.
And so that's exactly what these sections do.
So those sections are the things that are outputted by the policy part.
And then the inputs to the policy part are the outputs produced by the p system.

And so as I say, it's exactly dual.
So the outputs are the inputs that p needs
and the inputs are the things that p produces.
And so we can unravel this in the case of this monomial interface,
which is going to give us what we get in classical active inference.
The output path of one of these systems is going to do action selection.
So it has a little state space, which I just called R here.
It takes an observation and it produces an action.
And this is exactly what the sort of action selection process
and active inference does.
It takes a state and an observation and produces an action
that is to be taken in the generative model.
And it also has this part which does state inference
or you might see it called the recognition model
or it computes the Bayesian posterior over the states.
So we're in a state and we have an observation
and then we get a new state.
So these sort of four things,
these two parts of the policy system
and then the transition model
and the likelihood in the generative model
are basically all of what an active inference system is.
You might include the prior in that as well, I suppose.
And so that means that we've got all the data
of an active inference system in this very neat little package.
It's a generative model and it's dual and that's kind of it.
There are obviously some ways you can extend this
and that's not something I'm doing in this paper
or in this talk,
but basically you can do things like consider trajectories
over actions or continuous time models
or do lots of other things like this.
But I'm going to try to keep it as simple as I can.
I know I've introduced a lot of mathematical detail so far.
And then, okay,
so then there are three further generalizations
that kind of crop up pretty immediately from this.
that I think are quite important

and I want to just introduce to you all.
The first of them is that now we've got this home structure
and this nice two category.
We can think about, as I've indicated already,
think about hierarchical deep systems.
and that really corresponds to nesting small systems
and bigger systems.
And that formally means going from generative models
with trivial inner boundary
to generative models
which have a sort of non-trivial inner boundary.
And now, unfortunately,
I've swapped my direction of my home.
So it goes from Q to P.
The story still holds.
I've just changed P and Q round.
And I'll say in a moment
how this unwinds
so you can get a sense of what it does.
And then we do the same to the policy.
It just does the same thing
in the opposite direction.
Now it's a system from P to Q.
But they're still like dual.
So the intuitions are still good, I think.
That's one thing we can do.
This produces a sort of bi-category,
a two-category,
or what ultimately becomes a double-category
where the arrows are pairs of a system and a controller.
And so the next thing we can do to this
is to think about how do we model
in a sort of rigorous way
the changes of model,
changes of interface
that I introduced at the beginning.
That corresponds to adding
yet another dimension to this structure.

And that formally encodes changes of model
as extra morphisms categorically.
And that makes them kind of accessible
to this formalism.
Not something that I'm going to talk about today
because I've introduced enough stuff here
just to sort of whet your appetite, I think.
But in this formalism,
even in this sort of just
sort of two-dimensional setting,
because it's quite general,
because it says that active inference
is really made up of two key ingredients,
a system and a controller,
it allows us to ask precisely
how active inference relates
to other theories of agency
or other notions of system.
And of course,
lots of people have thought about
the relationship between active inference
and other accounts
like reinforcement learning.
But I think this framework
allows us to make those relationships
very, very precise indeed.
And to say, okay, well,
there is, say,
one might hope,
a particular,
like possibly universal way
of mapping any reinforcement learning system
into an active inference system
in a compositional way.
So that is,
you could do it for the parts
and also for the complex system,
and you get something that makes sense.

And possibly one could go,
and I think it's quite likely
that one could also go
in the other direction,
and there would be nice sort of,
we might expect
like nice formal results
like these two processes
were adjoint
or something like that.
But I would like to highlight
at this point
that active inference,
I think,
in this setting
is really not just
one theory of agency,
but really it's a whole family
of theories of agency
parameterized by things
like whether or not
you take continuous time
or discrete time systems,
because those,
of course,
give you different systems theories,
or whether you do planning
by sophisticated inference
or inductive inference
or just one time step,
or whether you,
or whether you think about
like multiple time steps,
or whether you take
the expected free energy
or the free energy
of the expected future.
So there are all

these different variations
and they give you
like different like notions
of active inference
and so probably different theories
of agency in this sense.
And I think that's important
for like making sense
of the whole active inference landscape
and in particular,
the sort of landscape
of theories of agency itself,
because it means
that we can have
like a nice rigorous map
of all of these things
and like connect them
from, you know,
from, you know,
connect one thing
into another quite nicely.
Again, I'm not going to say
very much about this,
but I just wanted
to flag it up,
but that's something
that you get out of
making precise these structures.
So I mentioned
that I wanted to highlight
how this captures
D for hierarchical
active inference.
And this happens
in, I think,
quite a natural way.
I mean,
I've introduced

an awful lot of formalism here,
but once you get used to it,
I think it works okay.
But hierarchical,
active inference
just corresponds
to a sequence
of composite arrows
in this two category
whose arrows
are made up of,
you know,
a generative model
and a controller
actually is upside down,
but it doesn't matter.
So in this setting,
again,
as I've mentioned,
the monomial interfaces
correspond to the
kind of normal marker
blankets that are just
made up of a set
of observations
and a set of actions,
but we get this
hierarchical structure
in the one cells
of this two category.
And if we think
of the sort of
composition in this two category
as basically by nesting,
then, you know,
we could,
I sort of draw it
schematically like this.

I'm indexing
like the higher levels
of the sort of interior
levels of the system
as like higher
in this context.
So like I plus one
is kind of more
inside the system
than I.
Often that corresponds
to like a higher level
of like explanation
in active inference,
but because there's
this duality
between like
the genitive model
and the controller
or the policy part,
I don't like to say
high level or low level
because it gets confusing.
I prefer to sort of
think about
how deep you are
inside your agent
and how sort of,
or how close you are
to the exterior world
when I'm talking
about the direction
of this kind of
hierarchy.
And so to the left
we go sort of more interior
and to the right
we go more exterior.

And these areas point
sort of in the exterior
direction like this.
And so a sort of
hierarchical active
inference system
goes from interior
to exterior.
It's a sort of,
it's an arrow
of this kind
that unwinds
to the following data.
It's a state F,
a likelihood
that is, you know,
parameterized by the state
and the interior
belief about observations.
So which you could think
of as like
high level
in some contexts,
observations.
And it produces
a prediction
about the more
exterior observations.
There's a prior
on the more interior
actions,
which again,
you could possibly
think of as
high level,
that this is parameterized
by the,
or conditioned

on the state
and the,
and the sort of
interior observations
actions and the exterior,
sort of more exterior
actions,
or the system's belief
about the more exterior
actions.
And then a transition
model,
which again,
depends on the actions
taken at this kind
of exterior level,
sort of high level,
in this case,
this is the sort of high
level actions,
and it gives us
a new state
on the basis
of the old state.
And then there's
this other part,
which is dual,
which is the controller.
So the control,
here I've written
that the state space
is like beliefs
over the,
over the generative model
state space.
So that says,
it takes,
it has this,

again,
these three parts.
So one is like
a generative process,
which takes the,
in the sort of
exterior observations,
the signals
which impinge
on the sort of
outer Markov blanket
and translates them
onto the inner
Markov blanket.
So it goes
in the other direction
like this.
There's a sort of
policy part,
which does action selection.
So it takes in
a belief over
the state space
and like a high level
or a more interior
belief about
over the actions
and it produces
what we could think
of as more
exterior choice
of actions.
So that could be
like passing down
a sort of
high level motor
signal into
like low level

sequence of like
muscle contractions
because I think
of the sort of
closer to the
exterior systems
as like the ones
being made of
lots of say
cells rather
than the,
the,
the interior
model of myself
as just a body
with arms that
move.
And then finally
there's this
component which
does,
Bayesian inference
over the states,
it takes in a state
and an exterior
observation,
that is the signals
which they impinge
on my retina
and it updates
the internal state.
And that's,
you know,
the part that does
like,
you know,
that gives you
your posterior

over states.
this is a lot of data.
In the literature
it sometimes
comes out
in a slightly
less
or maybe
slightly more
familiar way
or a slightly
less detailed way
where the,
you don't have
the interior
observations,
which I think
was like sort of
beliefs about
high level
patches of
information,
like the object
rather than the
sort of individual
like retinal
ganglion data
or something
like that.
And if you
imagine not
having these
OIs,
OI plus ones
in this model,
then you wouldn't
have like this
generative process

and you would
only have a
likelihood which
takes a state
to an observation,
but you still get
this like hierarchical
action selection
process.
So it just takes
like high level
action,
like high level
beliefs about
sort of actions
like I'm moving
my arm into
sort of lower
level sort of
signals that I
pass to my
muscles.
But I think
it's nice to
allow for this
hierarchy in the
opposite direction
on the
observations as
well.
Okay, so the
final thing I
want to mention
is that the
categorical systems
theory lets us
talk about,
lets us use

categorical logic
for talking
about goals
and behaviors
of systems.
And formally,
it's quite easy
to say what we
need to do
that, it's
not something
that I've done
because actually
there's a lot
of details,
but formally
like it's
relatively simple,
all you need
is the ability
to define
what's known
as the
fibrations.
And a
fibrations is
just like a
collection of
things indexed
by the other
things.
So again,
the sort of
index category
notion that I
introduced at the
beginning is
basically equivalent

to fibrations.
But the idea
is now that
over each
system,
we have
a fibrations
of predicates
of logical
statements
about those
systems that
is indexed
by the
systems and
the systems
indexed by
interfaces.
And it's
compatible with
all the changes
of system and
changes of
interface and
things like
that.
But one
notion,
one sort of
simple idea
of like logical
structure or
predicate you
might attach to
systems is just
notion of goal
for a system.
So these priors

on the state
space.
And I
haven't
elaborated this,
but once you've
got this,
we can kind of
use the
structures of
category theory
to do things
like define
quantifiers or
logical connectives
and sort of
build up
statements about
goals out of
sort of more
primitive statements
and compare
systems goals
and to say
possibly things
like, oh,
now if I've
got a bunch of
systems connected
together,
maybe they
work together
to achieve
their goal
possibly like
by some
process of
consensus.

This allows
us to talk
about that
kind of thing
in a mathematically
rigorous way.
Or now
we've got
this formal
logic to
talk about
the goals
of systems.
Can we use
this to
talk about
notions of
safety or
to guarantee
that the
goals will
be achieved
within certain
bounds or
that they
won't try
to do
this thing
or this
other thing
which we
don't like.
I think
this is
important
again
as I
mentioned

at the
beginning
for thinking
about
safe
artificial
systems.
That's all
I wanted
to talk
about.
It was
a lot.
I think
we can go
back and
elaborate on
any of
this.
And I'm
sure we'll
come back
to it in
the future
as well.
Thank you
so much
for having
me.
I hope
it wasn't
too
bamboozling.
But I'm
always willing
to answer
questions.
If people

have more
questions,
please feel
free to
get in
touch
afterwards
as well.
Thank you,
Toby.
Wow.
Okay.
Yeah.
While people
are asking
some questions
in the
live chat
which I'll
read to
you,
could you
reiterate,
how does
what you
refer to as
classic
active
inference
with
Markov
blanket,
Bayesian
graphical
formalism,
how does
that
overall

literature
relate to
what you
discussed
today in
terms of
applied
category
theory?
Right.
Okay.
So I
think the
very concise
way of
putting it,
let me see
if I can
go back
to a
slide
where it's
relatively
or not
too
intimidating.
The
concise
way of
putting it
is that
the
systems
in active
inference
that I
call
classical

that we're
more familiar
with
heretofore
are
basically
like
sub
theory
or a
sub
category
of this
whole
structure
that is
spanned
we would
say
in
mathematics
by these
monomial
interfaces.
These are
the ones
that don't
have this
complicated
state
dependence
on the
actions.
They just
have a
space of
possible
observations

and a
space of
actions.
And in
particular,
if you
don't think
about these
hierarchical
or deep
systems,
then it's
relatively simple
to see that
those systems,
those kind of
generative models
are captured
by this
formalism
because you
get
over one
of these
monomials,
the monomial
represents the
Markov blanket,
the
coefficient
here is
the
observations,
the
exponent
is the
actions,
we have a

state space,
we have a
prior,
we have a
likelihood,
and we have
a transition
model.
And that's
kind of the
whole deal
with a
generative
model.
The
policy part,
the thing
that I
call the
controller,
it kind
of remains
like I
said,
it's the
thing that's
dual,
it does,
let's see if I
can go back
to the
slide,
it does,
yeah,
it does
action
selection,
i.e.

planning,
and it
does base
and inference,
i.e.
it produces
the posterior.
Here,
R is just
like a
placeholder,
usually this,
at least in
the story
that I've
told in
the paper,
this would
be like
beliefs
over states,
D here
means
distributions.
So this
is distributions
over states.
It's just
aware of
packaging all
of that
up into
a nice
compositional
framework,
so that
we can,
because you

see a lot
of,
if you
look at
complicated
hierarchical
active
inference,
pictures,
they often
have
really
complicated
structure,
but actually
these structures
typically have
some kind
of common
repeating
elements,
that in
principle
should just
be able
to be
boxed up
and repeated
and plugged
in in a
nice way
that is
mathematically
sound and
reusable
and kind
of,
you know,

I like
category
theory
because it's
a kind
of
lingua
franca
and it
means that
you don't
have
small
variations
in
formalism
like
what does
message
passing
mean?
Well,
it means
lots of
different
things,
lots of
different
people.
In this
setting,
you can be
very precise
about what
your theory
of agency
is,
what the

notion of
Markov
blanket
is.
One of
the things
I often
find a bit
tricky
about the
active
inference
literature
is the
notion
of
Markov
blanket
is
intuitively
quite easy
to graph,
but in
this kind
of
dynamical
setting,
it doesn't
quite
correspond
to the
notion of
Markov
blanket
that
originally
came out
of

statistics.
Unless you
do some
kind of
maneuvers
that
unwind
your
dynamics
into
a big
graphical
model
and
draw
certain
boxes
around
some
repeating
elements,
then you
can say
that the
Markov
blanket
of the
system
is the
parents,
children,
co-parents,
like you
would say
in statistics.
But it's
kind of
not really

what the
job of
the Markov
blanket
is.
Not in
active
inference
and not
in
statistics.
They
play
different
roles.
In
active
inference,
the
Markov
blanket
is
the
agent
interface.
It's
like
the
boundary
of
the
agent
how
it
interacts
with
the
world.

I
think
it's
nice
to
be
precise
about
that.
It's
nice
to
say
that
when we
say
Markov
blanket
we
mean
interface
and
our
interfaces
come
together
and
they
can
connect
up
and
they
can
produce
bigger
systems
with

interfaces
and
that
has
a
particular
grammar
to
it
which
is
not
the
same
as
the
grammar
of
Markov
blankets
and
statistics
it
might
be
in
certain
cases
but
if
we
want
to
talk
about
the
whole
world

of
different
kinds
of
dynamical
systems
from
these
discrete
time
ones
that
I
was
talking
about
to
various
different
models
of
continuous
time
ones
then
it's
not
so
easy
to
make
the
connection
back
to
Markov
blanket
and

so
I
think
to
some
people
who
come
from
statistics
this
terminology
can
be
a
bit
confusing
and
one
of
how
we
can
be
precise
about
this
it
says
okay
well
we
have
this
notion
of
interface
it

can
be
whatever
you
want
as
long
as
it
behaves
like
we
think
of
as
interface
and
I
just
told
you
about
one
story
of
interfaces
and
basically
one
story
of
systems
over
the
interface
but
categorical
systems

theory
is
very
general
and
let
you
talk
about
lots
of
different
kinds
of
interface
lots
of
systems
and
this
is
why
I
think
that
this
story
of
structured
active
inference
whilst
it
captures
the
kinds
of
active

inference
systems
that
we're
familiar
with
it's
really
a
theory
of
the
structure
of
agents
generally
and
I
think
active
inference
wants
to
be
that
and
that's
why
I
think
that
here
we
might
want
to
expect
that

there's
like
canonical
ways
of
relating
other
theories
of
agency
to
active
inference
inference
and
that
we
can
make
those
mappings
precise
not
just
for
an
individual
model
or
an
individual
system
but
for
all
systems
of
a

particular
kind
and
that's
the
kind
of
universality
that
I
think
we
should
be
aiming
for
I
can
explain
what
these
arrows
on
this
page
mean
but
I
think
you
get
the
idea
that
this
is
to
say

if
you've
got
an
agent
it's
a
system
and
a
controller
you
can
forget
the
controller
and
you've
just
got
a
system
active
inference
is
a
way
of
saying
okay
given
my
system
I've
got
a
story
about

how
to
attach
controllers
to
it
how
to
do
policy
how
to
do
planning
how
to
do
action
selection
how
to
do
state
inference
active
inference
is
a
recipe
to
take
a
generative
model
and
provide
a
controller

and
reinforcement
learning
is
the
variance
other
direction
so
I
think
the
thing
to
take
away
is
that
you
know
if
you're
just
interested
in
simple
models
with
like
one
set
of
actions
and
one
set
of
observations

then
you
don't
have
to
worry
about
this
but
if
you're
interested
in
building
big
complex
models
of
big
complex
systems
like
organizations
or
if
you're
interested
in
talking
about
systems
whose
interfaces
can
be
dependent
or
whose

actions
might
that
themselves
are
like
dynamical
and
changing
then
we
need
stronger
mathematical
tools
for
describing
those
systems
and those
processes
and those
kinds
of
agents
and
that's
what
this
framework
and that's
what
categorical
systems
theory
gives
us
awesome

so
so many
great
points
and things
I'll
just
highlight
two
pieces
that I
think
really
hit
you
called
it
a
particular
grammar
that
knowingly
or
unknowingly
plays on
the
particular
partition
splitting
things
into
particles
and
being
that
partitioning
and
this

is
also
a
grammar
of
particles
and
it's
a
particular
grammar
like
there
are
sentences
that
are
syntactically
or
grammatically
correct
so that's
this
kind
also
there

is
this

there
is
this

conceptual
connection
that is
the
semantics
that
separates
from
the
strong
typing
of
the
syntax
so it
brings up
a lot
of
linguistic
and
then
also
you
said
it's
a
recipe
for
how
to
do
action
and
that
is
like
if

we
were
to
do
dynamical
systems
modeling
of
a
predator
prey
system
or
of
temperature
in a
room
but
then
to
bring
in
the
upstairs
control
section
can
open
up
multiple
possible
like
cascading
state
space
ramifications
especially
if you

allow
for
the
heading
in
and
out
of
the
nested
system
that
opens
up a
lot
of
dependencies
that
at
least
need
to
be
accounted
for
or
just
go
with
a
special
case
where
it's
nothing
other than
just
like

a
fixed
situation
which
is
kind
of
where
the
buck
stops
ultimately
for
any
hyper
hyper
hyper
prior
yeah
I
like
those
reflections
so I
think
yeah
there's
a
this
like
particles
idea
you know
I tried
to
capture
in
this

very
feeble
way
I'm
not
very
good at
drawing
here
you've
got
this
kind
of
nesting
of
systems
and
I'm
trying
to
capture
that
in
this
formalism
so
you
could think
of
these
outer
boundary
as being
like
your body's
boundary
or

boundary
or
the
boundary
of
a
whole
organization
or
society
or
ecosystem
and
then
the
sort
of
outer
like
the
job
of
the
outer
system
is
to
pass
messages
into
the
inner
parts
which
are
your
organs
or

units
of
an
organization
or
departments
or
something
like
that
or
lower
level
but
the
smaller
scale
or
interior
systems
one
thing
that's
not
so
easy
is
to
talk
about
the
birth
and
death
of
these
things
I haven't

really
thought
much
about
how
systems
are
created
from
others
but
I'm
trying
to
get
at least
some
framework
for
talking
about
what
it
means
to
be
one
before
we
talk
about
how
to
split
them
up
you
said

some
things
about
grammar
yes
and
I
think
it's
I'm
trying
to
use
this
very
sort
of
in
some
sense
quite
sparse
mathematical
grammar
of
category
theory
but I'm
trying to
do so
in a
way
that
is
quite
general
and
it

ends up
being
inherently
alien
in
but I
think
the
thing
you
gain
from
that
is
this
ability
to
talk
about
lots
of
different
kinds
of
things
and
hopefully
this is
just the
beginning
of this
project
that it
becomes
we get
tools and
stories for
how to

express
things in
this
kind of
language
it's
I think
most suited
for
trying to
understand
complex
systems
so
as I
say
I think
if you're
happy with
normal
active
inference
then it's
probably
fine
great
and also
I'm wearing
an appropriate
embroidered
onion
shirt
on the
birth and
death
even though
from each
nucleus's

or core's
perspective
it is
like being
at the
center of
a bullseye
from where
it is
taking
another
view
might look
more like
a chocolate
chip cookie
or like
multiple
nested
organs
so it's
not that
it's
like
just a
bullseye
even though
that's just
the two
layers
that's just
the motif
of nesting
and then
when you
described how
the systems
can be put

side by
side
then you
could have
a situation
where it's
like the
cookie has
two chocolate
chips and
then the
next time
step there's
a third
one but
it's
disconnected
so it's
the island
and then
another
kind of
actor comes
in and
changes the
wiring
secondarily
so one
thing I
wanted to
I haven't
really talked
about at
all but
one thing I
want to be
able to talk
about and

it's kind
of related
to this
question of
like when
the systems
dissolve into
other systems
or how can
we see
things as
like a
chocolate chip
cookie or a
bullseye or
something like
this or a
group of
trees or a
forest or
something
this thing
reduces
systems in
category
theory so
it turns
systems into
arrows
but really
this structure
wants to be
what's known
as a double
category so
here the
arrows would
be like the

changes of
interface and
then we would
maybe have
some other
structure down
here which
represents another
system and
then we have
like what are
known as
squares and
these squares
would represent
how we could
see like this
like this
system up
here as a
composite or
as a different
kind of system
how we could
see the forest
as trees or
something like
that so it
allows us to
represent these
kinds of like
dissolving or
these kinds of
changes I
haven't explained
in detail how
this works but
it's a step

towards that
kind of process
yeah cool and
then of course
all the great
juxtapositions with
more topologically
stable systems
neural networks
and then more
like the ant
nest mates okay
I'll ask a
question from
the live chat
math for
wisdom wrote
I wonder if
it is possible
to interpret it
in the opposite
way that the
policy is the
consciousness
so speculation
yeah what
what led you
to see or
identify one
with the other
yeah because
really I always
found that the
generative model
encapsulates
like the
world as I
perceive it

there's nothing
really that I
can do to
change like
we think of
so the
generative model
is a system
yeah so
there's one
peculiarity about
this which is
that
the key
ingredient is
like a
system whose
outputs are
observations
and that
seems a bit
weird sometimes
to me and
it may seem
weird to
people who
use
interactive
inference
where you
could think
of the
outputs of
an agent
should surely
be the
actions
but the

generative
model
the agent's
beliefs about
the world
is a
dynamical
system
which outputs
observations
that the
agent receives
and so
that seems
like the
wrong way
around
but it
is just
how the
mathematics
requires it
to be
and so
this thing
being so
crucial in
the structure
says to me
that this is
like the
sort of
this is
when we see
active inference
text is
often like
S

given O
and M
so it's
like conditioned
on a
particular
kind of
model
this
gamma
is M
and this M
doesn't typically
change in
active inference
unless it's
by some
process of
evolution
for an
individual
agent
this model
is fixed
and this
thing
represents
everything
that I
see
and it
represents
how I
set the
world to
change
given the
actions

I take
and it's
like it is
in some
sense
the agent's
world
in active
inference
there is
no
like
the agent
can't access
what the real
world is like
only has the
generative
model
and so that's
why I think
this is like
my world
this is like
my conscious
experience
and then
so what
actions I
take
I mean maybe
so maybe this
includes some
self-model
and I think
you know
I'm doing
planning

or I think
I'm you know
behaving in a
way that's
reasoned
but actually
I'm just
acting on
instinct
according to
these subconscious
processes
I mean I don't
know much
about consciousness
science
but I think
that that kind
of story
is possible
and so this
stuff seems
to be
the things
that
I don't
necessarily
directly
experience
in the same
way that I
directly
experience
like my
visual
signals
but they
are the

things that
make me
like me
they are the
things that
provide my
input to the
world
and again
that's all
like dynamics
and it's maybe
stuff that I
observe but
maybe I only
observe it
because it
forms like
an interior
part of this
model which
might have
some complex
structure of
the kind
that I was
talking about
so that's
why I
thought of
it that
way round
and of course
maybe there
is some
deliberative
planning process
or some

internal
reasoning that
goes on in
my conscious
experience and
I do have an
inner monologue
and often it
does seem like
that so I
think there is
some interplay
between these
but possibly
that stuff is
borne out by
having some
complex structure
in this rather
than in these
things that I
think of as
subconscious
processes
I don't
experience my
own Bayesian
inference
processes except
in the cases
that they go
wrong or I
have some
illusion
sometimes I'm
in a building
that's particularly
symmetrical and I

go out of a
lift and I go
into one place
and I'm on the
wrong side and it
all feels really
weird but that's
not the same
kind of
conflict that's
kind of an
unusual
experience
anyway that's
my feeling
there
what an
unusual
experience
I'll read
from math
for wisdom
he wrote
I think that
makes sense
and now I
understand the
distinction
better
the generative
model creates
beliefs
the policy
creates
actions
but our
actions are
fated

we are only
free to
choose our
beliefs
so our
consciousness
has to do
with the
beliefs we
take
our will
did I
understand
the
distinction
correctly
I think
so I
think that
sounds like
a good
way of
putting it
I mean
this is
this is
really just
my
basically
this is
like my
answer
philosophizing
I don't
know
I mean
it would
be nice

if this
matched
like
what people
think is
actually
going on
but it's
just what
the sort
of mathematics
seem to be
saying to
me at
least
towards the
applied
side
what
kinds
of
programming
languages
or
approaches
or
open source
software
or methods
can people
use to
start
exploring
this
that is
a great
question
and you

know what
at the
moment
that doesn't
have a
great answer
the best
answer
for now
is that
I mean
at least
for like
modeling
is that
in
Julia
Julia
the guys
at the
Topos
Institute
have been
producing
a collection
of packages
in the
sort of
milieu
of their
project
algebraic
Julia
which is all
about
representing
categorical
structures

and doing
compositional
modeling
in
Julia
but
Julia
itself
is not
like
necessarily
well suited
to
describing
things
describing
the
world
in this
particular
kind of
compositional
way
it has
its nice
compositional
features
but basically
in algebraic
Julia
they've
written
like a
little
sub
language
in
Julia

itself
that
encodes
these
categorical
structures
one of
the things
that's quite
nice about
category
theory
is that
it's
very
like
you know
it's about
composition
so it's
very modular
and it
gives you
a lot
of
things
that you
can
change
I'll
let me
go back
to where
I talked
about
systems
theory
right at

the very
beginning
yeah
so you
know
I just
write down
this arrow
and it
maps from
my category
of interfaces
to the
category
of
categories
and in
principle
I could
just
this should
be
like
a
little
program
which
takes
in
which
where
I
just
like
write
I
provide
it

with
a
polynomial
and it
gives
me
back
this
other
data
structure
this
category
of
systems
and
then
I
should
yeah
I
should
be
able
to
talk
about
more
systems
between
these
things
and
I
could
be
a
map

like
this
or
make
changes
to
this
and
have
my
computer
automatically
do
it
but
that
is
just
not
possible
with
any
kind
of
computational
tools
that
we
have
right
now
and
I
often
feel
that
it
should

be
possible
to
do
this
in
a
way
that's
not
necessarily
like
tied
to
a
single
programming
language
like
like
algebraic
Julia
is
to
Julia
there
are
also
like
dependently
type
programming
languages
for
proof
assistance
but
if

you
write
in
those
then
you
are
stuck
in
that
world
one
of
the
get
compiled
to
web
assembly
and
we
can
interact
in
a
sort
of
language
agnostic
way
and
this
this
mathematics
should
be
language
agnostic

but
we
don't
have
any
languages
or
tools
to
talk
about
it
yet
it's
something
that I
think
should
change
people
who are
interested
in
changing
it
should
talk
to
the
top
people
or
talk
to
me
and
I
would

like
to
see
tools
which
are
really
nicely
compositional
for
now
but
they're
not
like
optimal
thank
you
well
it's
an
optimal
agenda
so
that's
very
cool
and
to
be
able
to
connect
even
potentially
in
the
coming

months
between
the
algebraic
Julia
and
rx
infer
dot
JL
could
already
offer
some
cool
opportunities
yes
I
think
that
would
be
a
great
project
cool
um
over
the
last
years
like
working
on
this
from
your
dissertation

which
we
discussed
in
the
live
stream
54
series
all
the
other
fun
and
learnings
that
we've
had
um
how
have
you
seen
like
the
process
of
going
about
that
recipe
evolve
from
like
the
chapter
six
in

the
2020
textbook
how
do
you
feel
like
as
more
and
more
of
the
applied
category
theory
developments
are
formalized
and
then
also
like
operationalized
how
is
that
trajectory
of
like
what
it
looks
like
to
apply
active

inference
yeah
I think
it's
been
very
interesting
and very
exciting
I would
say
for a
long
time
I
was
also
like
quite
I guess
relatively
confused
about
things
in
some
way
and
this
is
something
that I
haven't
yet
been able
to
resolve
and

I'm
sure
some
people
know
how
it
could
be
resolved
but
I'm
just
not
one
of
them
which
is
that
active
inference
has
come
out
of
the
free
energy
principle
world
where
the
idea
is
that
actions
are

selected
in
the same
kind
of
way
that
you
do
perceptual
inference
by
minimizing
surprise
or free
energy
but
the
formal
structure
of
pure
perceptual
inference
is
it has
a similar
feeling
so it
has
something
that goes
from
inside
to
outside
and
then

something
that
goes
back
this
is
the
thing
that
does
prediction
and
then
this
is
the
thing
that
does
inference
and
so
the
structure
of
these
two
things
by no
surprise
obviously
has this
similar
bidirectional
feel
but I
haven't
yet

shown
that
this
action
selection
process
is
exactly
of the
same
type
as
basic
inference
for
perception
and I
would
like
it
to be
it's
kind
of
frustrating
to
me
that
so
I
thought
for
a long
time
how do
I
get
action

when
I
was
writing
my
thesis
how do
I
get
action
into
this
and
that's
what
this
project
really
has
been
about
getting
action
into
the
story
finally
but the
relationship
between
them
isn't
as neat
as I
would
like
and
so

if
anybody
wants
to
talk
about
that
I'm
always
interested
in
trying
to
show
that
action
selection
really
is
a
surprise
minimization
or
approximate
inference
so far
it's
only
vaguely
in
the
structure
and
this
reminds
me
of
something

I
wanted
to
say
in
response
to
your
remarks
at
the
end
when
I
just
that
I've
said
here
roughly
the
data
of
an
active
inference
is like
a
hierarchical
active
inference
system
it has
these
components
it's
got
like

a
likelihood
model
a
transition
model
a
policy
thing
a
state
inference
thing
and
maybe
it's
got
this
genitive
process
part
and
this
action
prior
part
as
well
but
this
is
all
just
structured
this
is
structured
active

inference
is
all
just
about
structure
none
of
this
yet
is
about
how
do
you
actually
choose
this
likelihood
or
how
do
you
actually
do
action
selection
or
how
do
you
actually
do
state
inference
the
work that I was doing
in my thesis

was about
various different ways
that we actually do
inference
particularly
inference in the brain
using the
variational inference
under the
Laplace
approximation
but
what
I
want
is
to
say
that
active
inference
or a
particular
flavor of
active
inference
is
in some
sense
universal
and that
means
there
is
a
canonical
choice
for a

given
system
so
we
assume
that
somebody
comes
along
and
gives
you
the
data
of
your
generative
model
it
could
be
you
Daniel
or
it
could
be
evolution
science
or
something
like
that
and
then
it
says
okay

well
active
inference
says
there's
a
canonical
way
of
choosing
actions
that
is
as
optimal
as
you
can
be
and
there's
a
canonical
way
of
doing
inference
that
is
as
optimal
as
you
can
be
and
this
story

that
I've
told
is
agnostic
to
that
a lot
of
people
have
spent
time
saying
here
are
proposals
this
framework
although
being
agnostic
allows
you
to
state
what
are
known
in
category
theory
as
universal
properties
and
so
those

are
ways
of
saying
that
this
is
in
some
sense
canonical
or
optimal
in
some
mathematically
precise
way
and
it
would
be
nice
to
have
those
stories
about
the
universality
of
active
inference
i.e.
the ways
of spelling
out
these

functions
to be
expressed
in
that
language
so
yeah
that's
my
answer
two
answers
one
was
about
this
annoying
thing
with
inference
and
here
about
the
universality
yeah
there's
so much
to go
into
and you
mentioned
like
dual
a lot
and the
interface

can be
seen
it's
like
two
onions
where
you
could
peel
off
the
interface
and
associate
it
with
one
but
that
would
be
a
symmetry
break
because
you
could
also
see
it
as
being
peeled
off
with
the
other

because
it's
an
interface
and
this
gets
into
systems
engineering
what is
the
interface
and
it's
a very
interesting
topic
so
I'll see
if anyone
else has
any
final
questions
like
what are
your next
steps
on the
research
how do
you see
it being
applied
and
continued
okay

right
so
my
next
steps
are
really
to
produce
a
longer
or
more
detailed
paper
about
this
than
the
abstract
that
is
available
at the
moment
in
particular
the
abstract
that's
available
at the
moment
it
doesn't
really
spell
out

some
of
this
stuff
in
particularly
nice
detail
like
to
keep
things
as
simple
as
I
could
I
didn't
really
explain
like
how
you
can
have
this
kind
of
mode
dependence
transition
model
that
is
stochastic
as well
as

having
likelihoods
that are
stochastic
for technical
reasons
it's a lot
easier
if you
just
have
the
outputs
of your
systems
be
deterministic
this
notation
right
like
some
with
a
sampling
is
not
something
that
is
I
think
canonical
or
widely
known
so
I'd

have

to

spell

that

out

so

there

are

lots

of

mathematical

details

so

that's

the

first

thing

and the

second

thing

I think

is to

get people

using

this

framework

actually

talking

about

building

complex

systems

or

actually

building

tools

for

building
complex
systems
using
it
and
relating
to
the
tools
that
people
already
use
for
doing
that
kind
of
thing
like
Pi
MDP
or
just
like
you
said
Daniel
RX
and
Thur
it
would
be
great
to
actually

turn
this
into
something
that
people
can
use
rather
than
just
a
mathematical
theory
I
like
this
although
it's
kind
of
often
people
think
it's
quite
an
alien
language
because
it's
unfamiliar
but
it
is
in
some

ways
quite
ergonomic
it
is
quite
demanding
because
to
prove
that
something
is
an
index
category
you
have
to
check
all
of
these
annoying
conditions
and
by
God
they
are
often
really
annoying
and
I
would
love
as

a
test
case
to
build
an
artificial
proof
assistant
or
system
that
can
help
me
do
these
checks
because
as I
said
doing
policies
action
selection
is
very
much
like
choosing
a
series
of
morphisms
and
it's
well
known

in
mathematics
that
proof
is
also
choosing
a
series
of
morphisms
so
I'm
pretty
sure
that
we
can
use
this
for
doing
things
like
proof
in
a
way
that's
nicer
than
large
language
models
so
once
we've
got

tools
I
want
to
actually
start
applying
them
to
things
complicated
structured
models
that
are
probably
beyond
the
systems
that
we
have
right
now
!
okay
the
Toby
signal
has
gone
out
I'll
ask
one
more
question
from

the
chat
math
for
wisdom
wrote
Toby
it
sounds
like
you
want
to
set
up
a
free
functor
so
that
suggests
considering
the
adjoint
functor
what is
it
that
you
want
to
forget
if
he
could
define
a
functor

what
he
wants
to
forget
then
mathematically
that
could
yield
the
functor
that
freely
optimally
constructs
the
canonical
way
to
choose
the
action
or
do
the
inference
yeah
what do
I want
to
forget
in
some
sense
what I
want
to

forget
is
like
all
the
stuff
that
I
might
have
learned
that
confuses
me
but the
thing
is
it's
very
hard
to
distinguish

the
stuff
that
is
confusing
from
the
stuff
that
is
not
confusing
until
you've
done

all
the
work
so
it
may
be
the
case
that
you
have
to
construct
that
adjoint
in order
to see
that
you've
got
the
forgetful
funct
in
the
first
place
and
that's
unfortunately
I think
the
situation
that I'm
in
part
like

a lot
of
this
process
I
think
has
really
been
about
figuring
out
the
essence
of
these
systems
like
in
before
the
essence
of
the
structure
of
active
inference
really
distilling
it down
into
something
quite
abstract
and
really
just to

say
something
about
forgetful
functions
this
where has
it gone
yep
this arrow
here
that
goes
from
agents
the category
of agents
which has
these
bidirectional
genital
model
policy
things
as its
morphisms
and
just
goes
down
to
systems
this
is
a
forgetful
function
it

forgets
the
controller
the
thing
that
is
the
policy
and
possibly
hopefully
the
way to
see
active
inference
is
universal
is
to
say
that
it's
like
adjoint
in a
particular
way
or
at
least
it
has
like
at
least
it

is
a
section
we
would
say
of
this
thing
so
yeah
it's
true
what
I'm
trying
to
do
is
find
out
what
those
adjoints
are
but
it's
it's
hard
to
know
what
to
forget
until
you've
forgotten
it

and then
it's
gone
I don't
know
you
don't
know
what
you
got
till
it's
gone
then
there's
the
reconstructed
trace
and
our
niche
is
becoming
increasingly
capable
from a
perception
and action
standpoint
which
is bringing
up so
many
fascinating
topics
so
do you

have
any
final
comments
not
today
I'm
sure
there'll
be
other
opportunities
then
I
hope
so
I
do
it's
been
a
pleasure
thank
you
thank
you
toby
this
was
an
awesome
presentation
it's
like
big
structural
evolution
and development

in active
inference
and so
there's so
many pieces
for people
to
to pick
up
and
and work
in
always
developing
like the
background
needed to
get to
some
of these
cliffs
and
then
the
better
structures
to
jump
off
them
with
yeah
please
everybody
get in
touch
if you
have

questions
or
projects
in mind
I'm
always
interested
to
have
them
cool
and I
hope
that
we
can
like
reactivate
some
category
theory
act
in
the
rx
infer
project
or
just
in
some
other
standalone
project
I'm
sure
it'll
happen

so
thank
you
peace
toby
thanks
everybody
in
live
tsch
tsch
tsch
to

Thank you.